DATA ANALYTICS PROJECT REPORT

# Vehicle Fuel Consumption & CO2 Emissions

What really drives tailpipe emissions? An end-to-end analysis of 1,067 vehicles covering exploration, segmentation and machine-learning prediction.

**DATASET**
1,067 vehicles | 13 variables | Model year 2014

**TOOLS**
Python, Pandas, Matplotlib, scikit-learn

**TECHNIQUES**
EDA, correlation, K-Means clustering, Regression, Random Forest

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# At a Glance

Road transport is one of the largest sources of greenhouse gas emissions. This project asks a practical question: given a vehicle's basic specifications, how much CO2 will it emit, and which choices matter most?

| 1,067 | 256 g/km | 0.966 | 401 g/km |
|---|---|---|---|
| vehicles analysed | average CO2 emissions | cross-validated R2 of best model | passenger-van average, 1.85x a compact car |

## Project objectives

- **Profile** the dataset and verify its quality.
- **Explore** how emissions vary by vehicle class, fuel type, cylinders and brand.
- **Quantify** which specifications are most strongly linked to CO2.
- **Segment** the market into distinct vehicle groups using K-Means clustering.
- **Predict** CO2 from simple specifications and test how reliable the prediction is.

## Key findings

| | |
|---|---|
| 1 | **Fuel burned is the core driver.** Combined fuel consumption correlates 0.89 with CO2; each extra L/100 km adds about 16 g/km. Engine size adds about 39 g/km per litre. |
| 2 | **Emissions are concentrated.** The 235 vehicles above 300 g/km are 22% of the fleet but produce 30% of total emissions. |
| 3 | **Vehicle class decides the baseline.** Passenger vans average 401 g/km against 217 g/km for compact cars, an 85% gap. |
| 4 | **Transmission is a myth.** Manuals look 35 g/km cleaner overall, but within the same cylinder count the gap is under 5 g/km. |
| 5 | **Hybrids are the clear winners.** The 27 hybrid-named models average 165 g/km, 36% below the other vehicles (259 g/km). |
| 6 | **CO2 is highly predictable.** A Random Forest on just three inputs reaches R2 = 0.966 in 5-fold cross-validation, with a mean error of about 4 g/km. |

BUSINESS IMPACT If every vehicle above its own class median were brought down to that median, average fleet emissions would fall by about 8% (from 256 to 235 g/km). This is an illustrative scenario, not a forecast.

# 1. Data Profile & Quality

The dataset holds 1,067 vehicles from model year 2014, described by 13 variables: make, model, class, engine size, cylinders, transmission, fuel type, three fuel consumption measures (city, highway, combined in L/100 km), combined mpg, and CO2 emissions in g/km. The target variable is CO2 emissions.

| Dimension | Detail |
|---|---|
| Records / columns | 1,067 rows x 13 columns |
| Manufacturers / models | 39 makes (Ford largest with 90) / 663 distinct models |
| Vehicle classes | 16 classes (Mid-size largest with 178, then Compact 172, Small SUV 154) |
| Fuel types | Regular X (514), Premium Z (434), Ethanol E (92), Diesel D (27) |
| Engine / cylinders | 1.0 to 8.4 L / 3 to 12 cylinders (4, 6 and 8 cover 96% of vehicles) |
| Target (CO2) | 108 to 488 g/km, mean 256, median 251, std dev 63 |

## Data quality checks

| Check | Result | Action taken |
|---|---|---|
| Missing values | None in all 13 columns | No imputation needed |
| Duplicate rows | 0 | None |
| Constant column | MODELYEAR = 2014 for every row | Excluded from modelling |
| Outliers (IQR rule on CO2) | 6 vehicles | Kept: genuine heavy vans and luxury cars |
| Redundant column | Combined mpg is a unit conversion of L/100 km | Dropped from model inputs |
| Correlated predictors | City, highway and combined overlap strongly | Combined used as the single fuel input |

## Analysis workflow

**1 Profile**
Structure, types, missing values, duplicates

**2 Explore**
Distributions and group comparisons

**3 Relate**
Correlation and driver analysis

**4 Segment**
K-Means market clustering

**5 Predict**
Regression, ensembles, cross-validation

# 2. Exploratory Analysis

## 2.1 How much do vehicles emit?

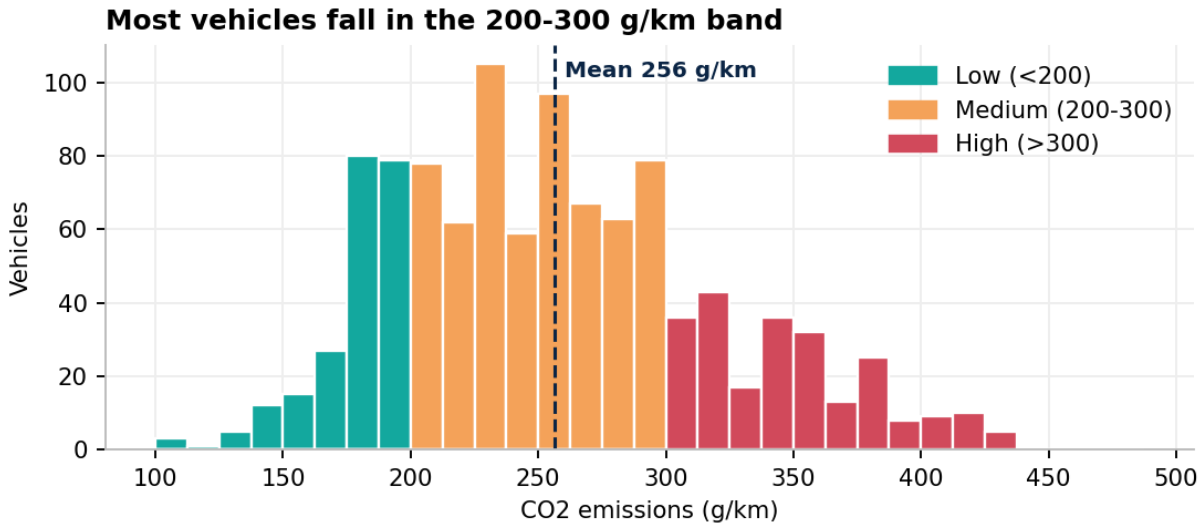



*Figure 1: Distribution of CO2 emissions, coloured by emission band*

To make the spread easy to communicate, vehicles were grouped into three bands: Low (under 200 g/km, 222 vehicles, 21%), Medium (200 to 300 g/km, 610 vehicles, 57%) and High (over 300 g/km, 235 vehicles, 22%). The distribution is mildly right-skewed: a tail of heavy vehicles stretches to 488 g/km.

## 2.2 Fuel type

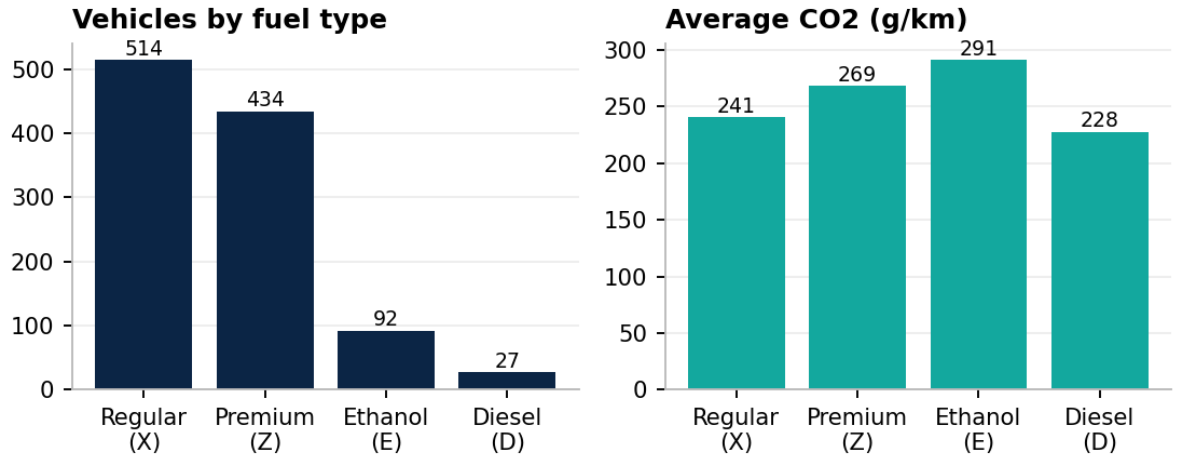



*Figure 2: Fuel type mix (left) and average CO2 by fuel type (right)*

Gasoline dominates (89% of vehicles). A useful check: dividing CO2 by fuel consumption gives a near-constant factor per fuel, about 23 g/km per L/100 km for gasoline, 27 for diesel and 16 for ethanol. This is simply the carbon content of each fuel, and it confirms the data is internally consistent. Ethanol vehicles show high average CO2 (291 g/km) because they are mostly large pickups and vans, not because the fuel itself is dirtier. Diesel has only 27 vehicles, so its average should be read with care.

## 2.3 Vehicle class

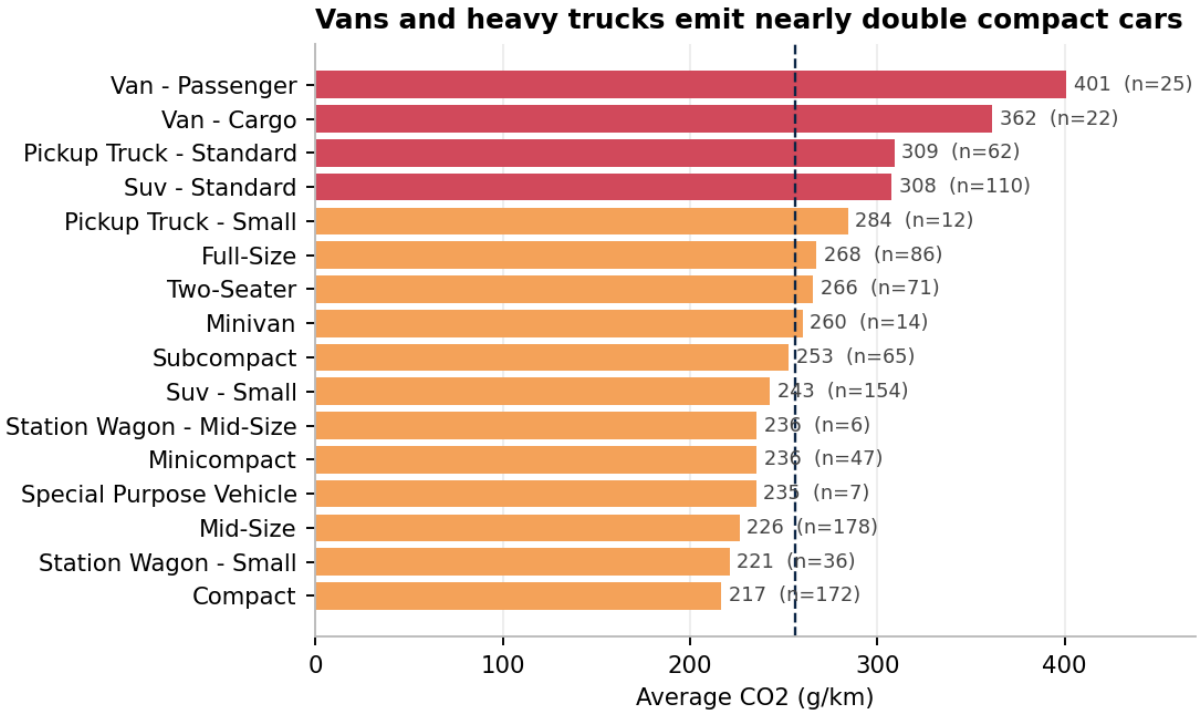



*Figure 3: Average CO2 by vehicle class (dashed line = overall mean of 256 g/km; n = vehicles)*

**INSIGHT** Compact (217), small wagon (221) and mid-size (227) cars are the cleanest classes. Standard SUVs (308) and pickups (309) sit well above the mean, and vans are far above it: cargo 362 and passenger 401 g/km. Class is essentially a proxy for vehicle weight.

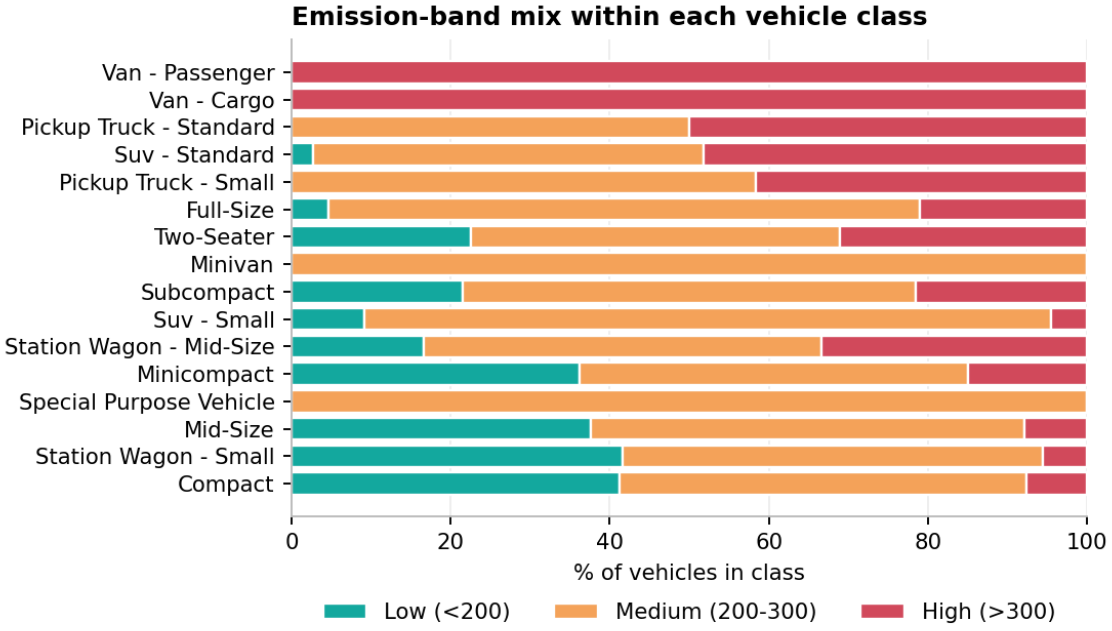



*Figure 4: Share of Low / Medium / High emitters inside each class*

The band mix shows how uneven classes are. Compact and mid-size cars supply most of the Low-band vehicles (71 and 67 of the 222), while every passenger van and nearly every cargo van sits in the High band.

# 3. What Drives CO2 Emissions?

## 3.1 Engine size and fuel consumption

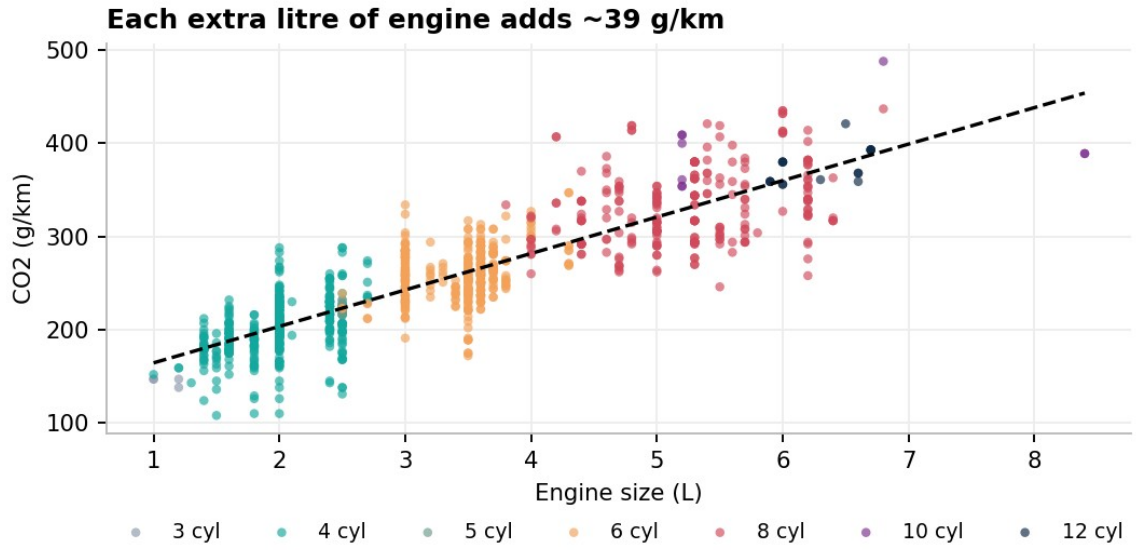



*Figure 5: Engine size vs CO2, coloured by cylinder count (dashed = linear trend)*

Emissions rise with engine size, and the cylinder colours show clean layers: 4-cylinder engines occupy the bottom of the cloud and 8-cylinder engines the top. Average CO2 steps up from 195 g/km for engines of 2 L or less, to 233 (2 to 3 L), 262 (3 to 4 L), 319 (4 to 5 L) and 348 g/km above 5 L.

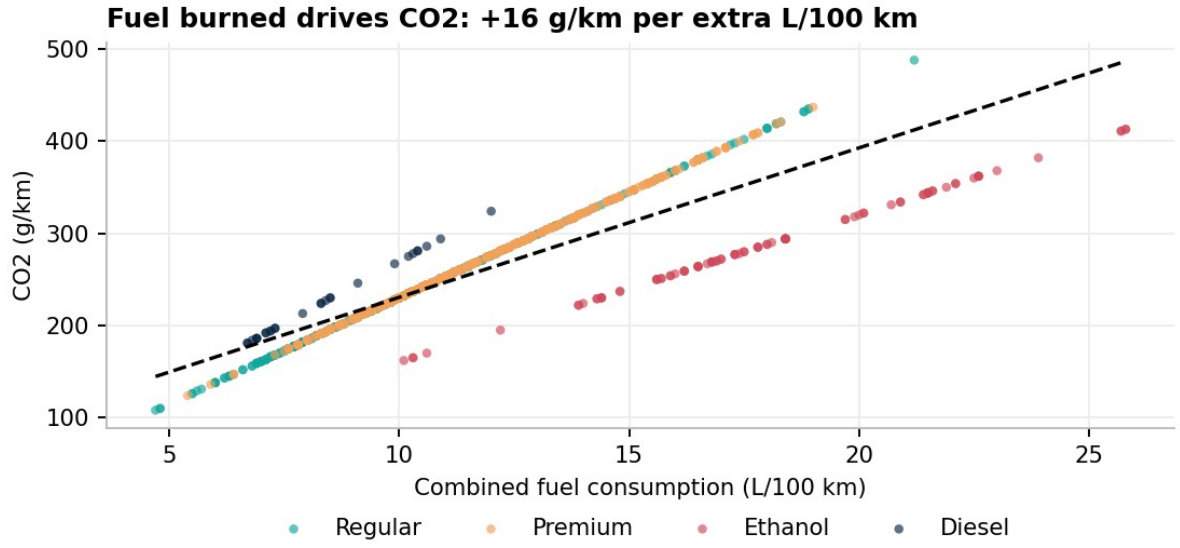



*Figure 6: Combined fuel consumption vs CO2, coloured by fuel type*

The fuel link is tighter and nearly linear because CO2 is a direct product of combustion. The remaining scatter is explained by fuel type: ethanol points sit lower on the CO2 axis for the same consumption, diesel points sit higher.

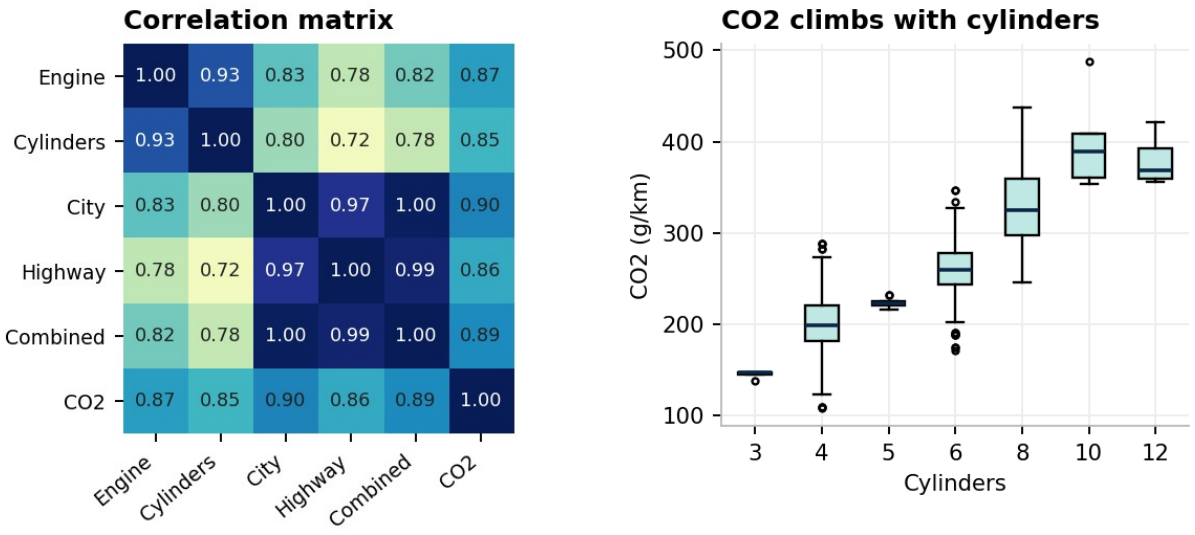



*Figure 7: Correlation matrix (left) and CO2 by cylinder count (right)*

Engine size (0.87), cylinders (0.85) and every fuel measure (0.86 to 0.90) all correlate strongly with CO2, and strongly with each other. This multicollinearity is harmless for prediction but means individual coefficients should not be over-interpreted. Moving from 4 to 8 cylinders raises average CO2 from 201 to 332 g/km, about 65% more. City consumption averages 1.4x the highway figure.

# 4. Vehicle Segmentation (K-Means)

Instead of imposing categories, K-Means clustering (k = 4, standardised engine size, cylinders, fuel consumption and CO2) was used to let the data reveal natural vehicle groups.

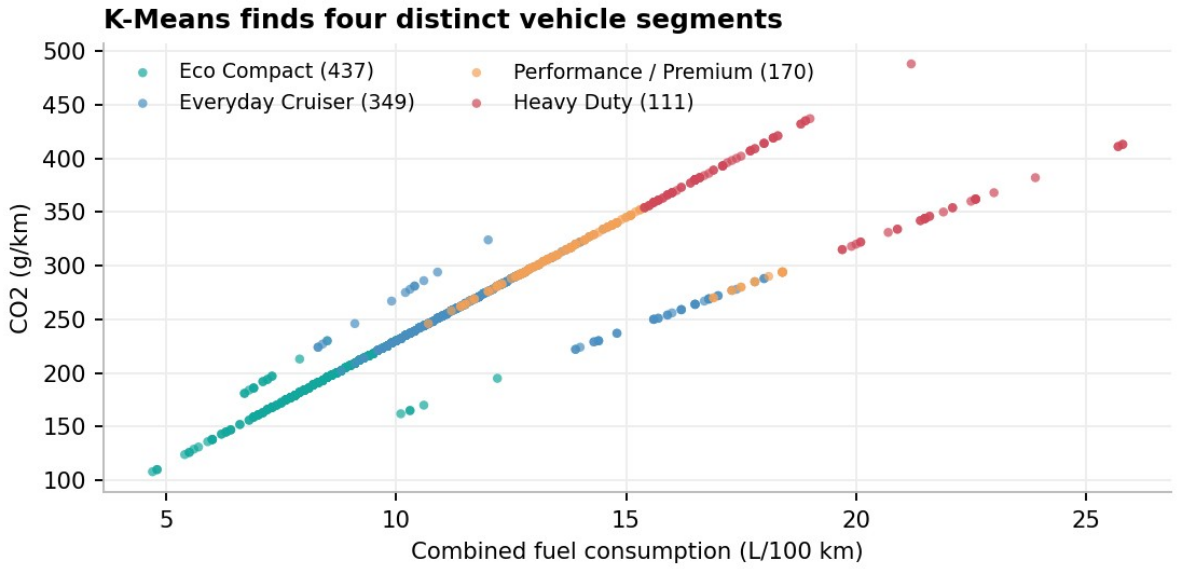



*Figure 8: Four data-driven vehicle segments*

| Segment | Vehicles | Engine (L) | Cylinders | L/100 km | CO2 (g/km) | Most common classes |
|---|---|---|---|---|---|---|
| Eco Compact | 437 | 2.0 | 4.1 | 8.7 | 200 | Compact, Mid-Size |
| Everyday Cruiser | 349 | 3.4 | 6.0 | 11.9 | 263 | SUV - Small, Mid-Size |
| Performance / Premium | 170 | 5.1 | 8.0 | 14.0 | 310 | SUV - Standard, Full-Size |
| Heavy Duty | 111 | 5.7 | 8.8 | 18.4 | 377 | Van - Passenger, SUV - Standard |

**INSIGHT** The Heavy Duty segment is only 10% of vehicles (111) yet averages 377 g/km, almost double the Eco Compact group (200 g/km, 437 vehicles). Targeting a small, identifiable group of large vehicles gives the biggest emission payoff.

### Brands and transmissions

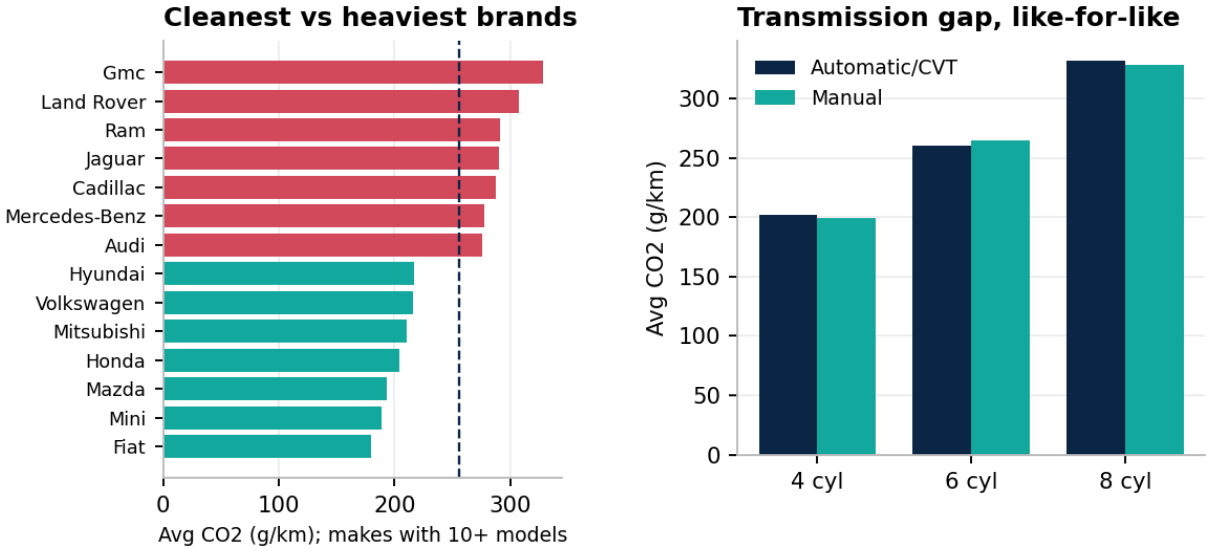



*Figure 9: Brand averages (left) and CO2 by transmission within the same cylinder count (right)*

Brand averages mostly reflect product mix: Fiat (180), Mini (189) and Mazda (193) build small cars, while GMC (328), Land Rover (308), Ram and Jaguar (291 each) sell trucks and large luxury vehicles. On transmission, manual cars average 228 g/km against 263 for automatics, but that is a mix effect. Like for like, 4-cylinder automatics average 202 g/km against 199 for manuals, and 8-cylinder cars are 332 against 329. Transmission type itself has almost no effect.

# 5. Predictive Modelling

Three inputs (engine size, cylinders, combined fuel consumption) were used to predict CO2. Data was split 80/20 (853 train, 214 test, seed 42), and every model was also checked with 5-fold cross-validation so the score does not depend on one lucky split.

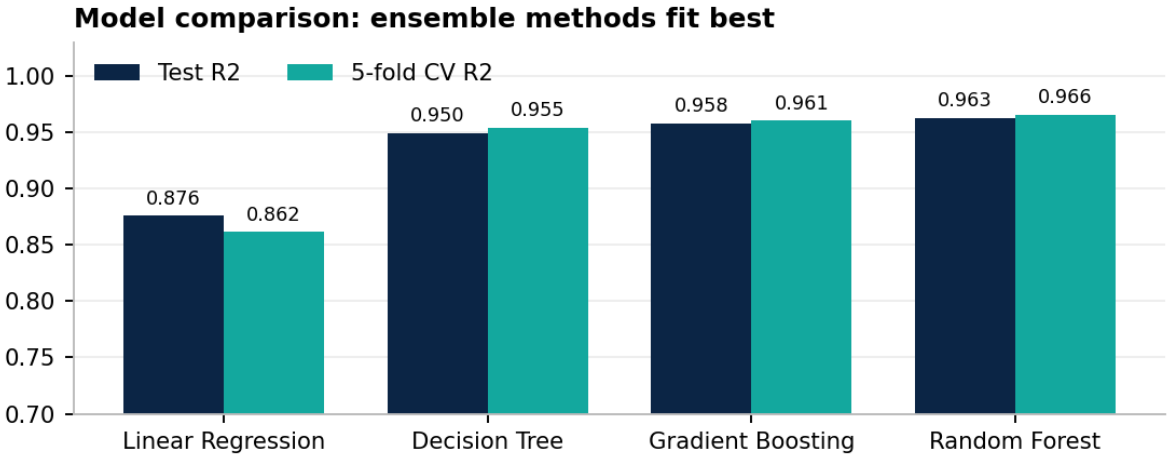



*Figure 10: Test R2 and 5-fold cross-validated R2 for four models*

| Model | Test R2 | CV R2 (mean +/- sd) | MAE (g/km) | RMSE (g/km) |
|---|---|---|---|---|
| Linear Regression | 0.876 | 0.862 +/- 0.013 | 16.7 | 22.6 |
| Decision Tree | 0.950 | 0.955 +/- 0.011 | 4.2 | 14.4 |
| Gradient Boosting | 0.958 | 0.961 +/- 0.006 | 6.8 | 13.1 |
| Random Forest | 0.963 | 0.966 +/- 0.005 | 4.4 | 12.3 |

Linear regression explains 88% of the variance, a solid baseline. Tree-based models capture non-linear effects and interactions and jump to about 96%. Random Forest is the most accurate and also the most stable across folds (standard deviation 0.005). Its mean absolute error of 4.4 g/km is under 2% of the average emission.

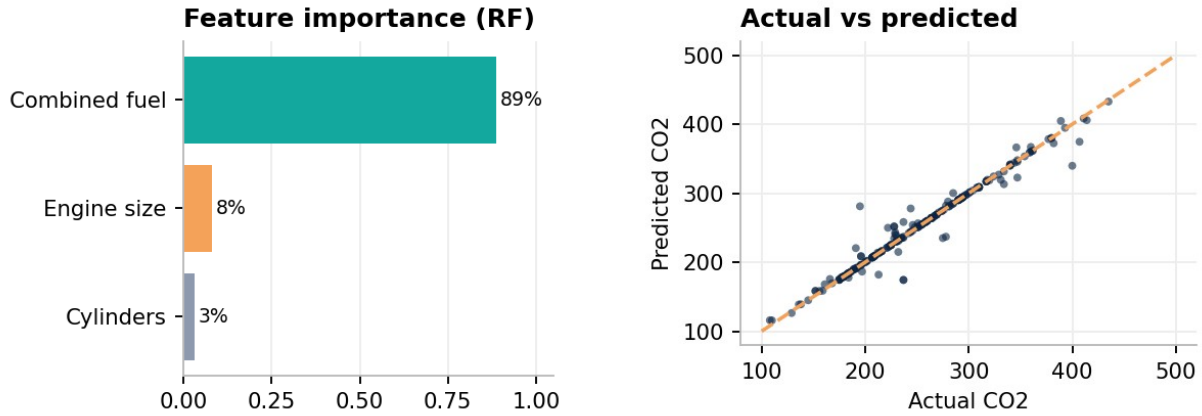



*Figure 11: Feature importance (left) and actual vs predicted CO2 on the test set (right)*

Combined fuel consumption carries 89% of the importance, engine size 8% and cylinders 3%. Once fuel use is known, engine specifications add little. Residuals are centred on zero (mean -0.01) with a standard deviation of 12.4 g/km; the largest single miss is 86 g/km, on a rare high-emission vehicle.

**Baseline equation (linear model):** CO2 = 67.3 + 11.21 x Engine + 7.16 x Cylinders + 9.52 x Fuel

# 6. Insights & Recommendations

## Best and worst vehicles in the dataset

**Lowest emitters**

| Make | Model | Class | CO2 (g/km) | L/100 km |
|---|---|---|---|---|
| Toyota | Prius C | Compact | 108 | 4.7 |
| Honda | Accord Hybrid | Mid-Size | 110 | 4.8 |
| Toyota | Prius | Mid-Size | 110 | 4.8 |
| Volkswagen | Jetta Turbo Hybrid | Compact | 124 | 5.4 |
| Ford | Fusion Hybrid | Mid-Size | 126 | 5.5 |
| Toyota | Prius V | Station Wagon - Mid-Size | 126 | 5.5 |

**Highest emitters**

| Make | Model | Class | CO2 (g/km) | L/100 km |
|---|---|---|---|---|
| Ford | E350 Wagon | Van - Passenger | 488 | 21.2 |
| Bentley | Mulsanne | Mid-Size | 437 | 19.0 |
| Chevrolet | Express 3500 Passenger | Van - Passenger | 435 | 18.9 |
| Gmc | Savana 3500 Passenger | Van - Passenger | 435 | 18.9 |
| Chevrolet | Express 2500 Passenger | Van - Passenger | 432 | 18.8 |
| Gmc | Savana 2500 Passenger | Van - Passenger | 432 | 18.8 |

Every one of the six cleanest vehicles is a hybrid, and five of the six dirtiest are full-size passenger vans.

### Recommendations

| | |
|---|---|
| **Consumers** | Compare vehicles on L/100 km and CO2 g/km, not engine size alone. Moving from a standard SUV (308 g/km) to a mid-size car (227 g/km) cuts expected emissions by about 26%. |
| **Manufacturers** | Prioritise hybridisation and downsizing in vans, pickups and standard SUVs, the segments with the highest baseline and the most room to improve. |
| **Policy makers** | Target the heavy tail: the 22% of vehicles above 300 g/km generate 30% of emissions. Class-based efficiency standards would reach them directly. |
| **Analysts / dealers** | Use the three-input Random Forest to estimate CO2 within about 4 g/km when full test data is not yet available. |

# 7. Limitations & Next Steps

### Limitations

- Single model year (2014): no trend analysis, and no modern hybrids, plug-ins or electric vehicles.
- Ratings are standardised test figures and can differ from real-world driving.
- Vehicle weight, aerodynamics and turbocharging are not in the data, though they affect emissions.
- Small groups (diesel 27, special-purpose 7, mid-size wagons 6) give less reliable averages.
- Hybrids were identified from model names, so a few may be missed.
- Correlation is not causation, as the transmission example shows.

### Next steps

- Add more model years to measure how efficiency has improved over time.
- Include weight and powertrain data and tune models with grid search.
- Publish an interactive Power BI / Tableau dashboard so non-technical users can explore the segments.
- Deploy the model as a small web app that estimates CO2 from three inputs.

# 8. Conclusion

CO2 emissions are driven mainly by how much fuel a vehicle burns, which follows engine size, cylinder count and class. A small group of large vehicles, 22% of the fleet, creates 30% of emissions, while hybrids show a 36% reduction against the rest. Because three simple specifications predict CO2 with R2 of 0.966, quick and transparent screening of vehicles is realistic, and efficiency policy can be aimed precisely where the impact is largest.

# Appendix: Column Reference

| Column | Description | Unit / type |
|---|---|---|
| MODELYEAR | Model year of the vehicle | Year (all 2014) |
| MAKE | Manufacturer | Text (39) |
| MODEL | Vehicle model name | Text (663) |
| VEHICLECLASS | Body style / size class | Text (16) |
| ENGINESIZE | Engine displacement | Litres |
| CYLINDERS | Number of cylinders | Count |
| TRANSMISSION | A automatic, AS auto-select, AV CVT, M manual, plus gear count | Text (22) |
| FUELTYPE | X regular, Z premium, E ethanol, D diesel | Text (4) |
| FUELCONSUMPTION_CITY | City fuel consumption | L/100 km |
| FUELCONSUMPTION_HWY | Highway fuel consumption | L/100 km |
| FUELCONSUMPTION_COMB | Combined fuel consumption | L/100 km |
| FUELCONSUMPTION_COMB_MPG | Combined fuel economy | Miles per gallon |
| CO2EMISSIONS | Tailpipe CO2 emissions (target variable) | g/km |

## Methods summary

- Emission bands: Low < 200, Medium 200 to 300, High > 300 g/km.
- Clustering: K-Means, k = 4, n_init = 10, standardised features, random_state = 42.
- Models: scikit-learn Linear Regression, Decision Tree (max depth 8), Gradient Boosting, Random Forest (300 trees).
- Validation: 80/20 hold-out plus 5-fold shuffled cross-validation (seed 42).